



Parktown Boys' High School

Geography Department

Grade 12 Control Test

Examiner: Mr A Meintjes

Time: 60 Minutes

Moderator: Mr D Grace

Total: 60 Marks

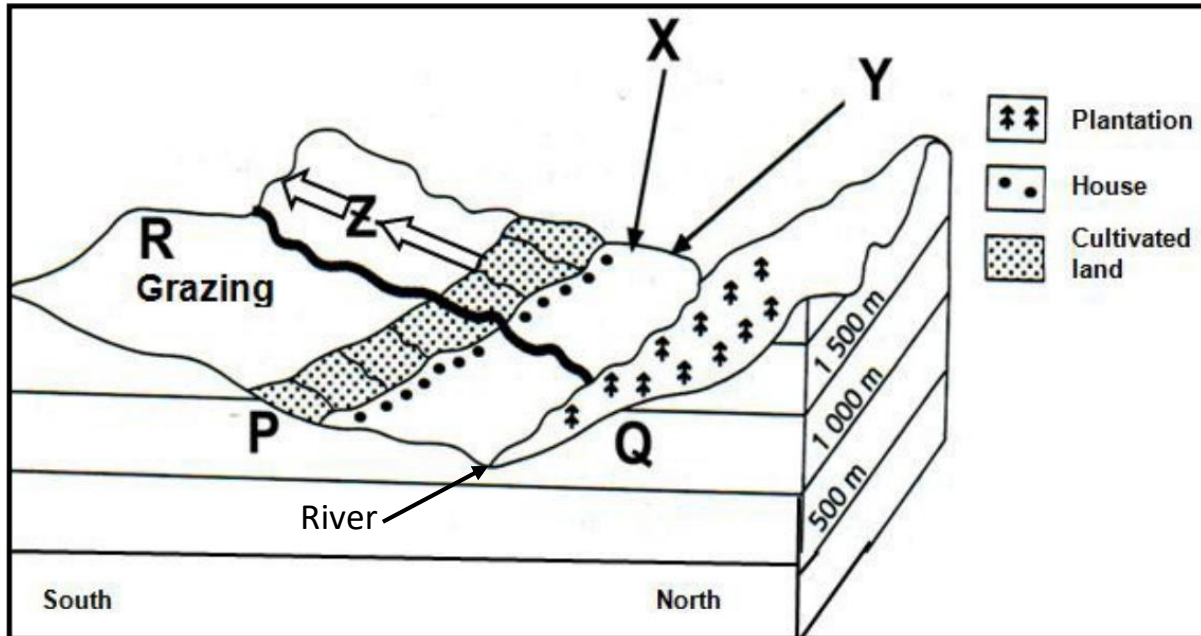
Date: 26 February

INSTRUCTIONS AND INFORMATION

1. The question paper consists of FOUR questions.
2. ALL diagrams are included in the question paper.
3. Where possible, illustrate your answers with labelled diagrams.
4. Leave a line between subsections answered.
5. Number your answers correctly according to the numbering system used in this question paper.
6. Write neatly and legibly.

Question 1 – Short Questions [15]

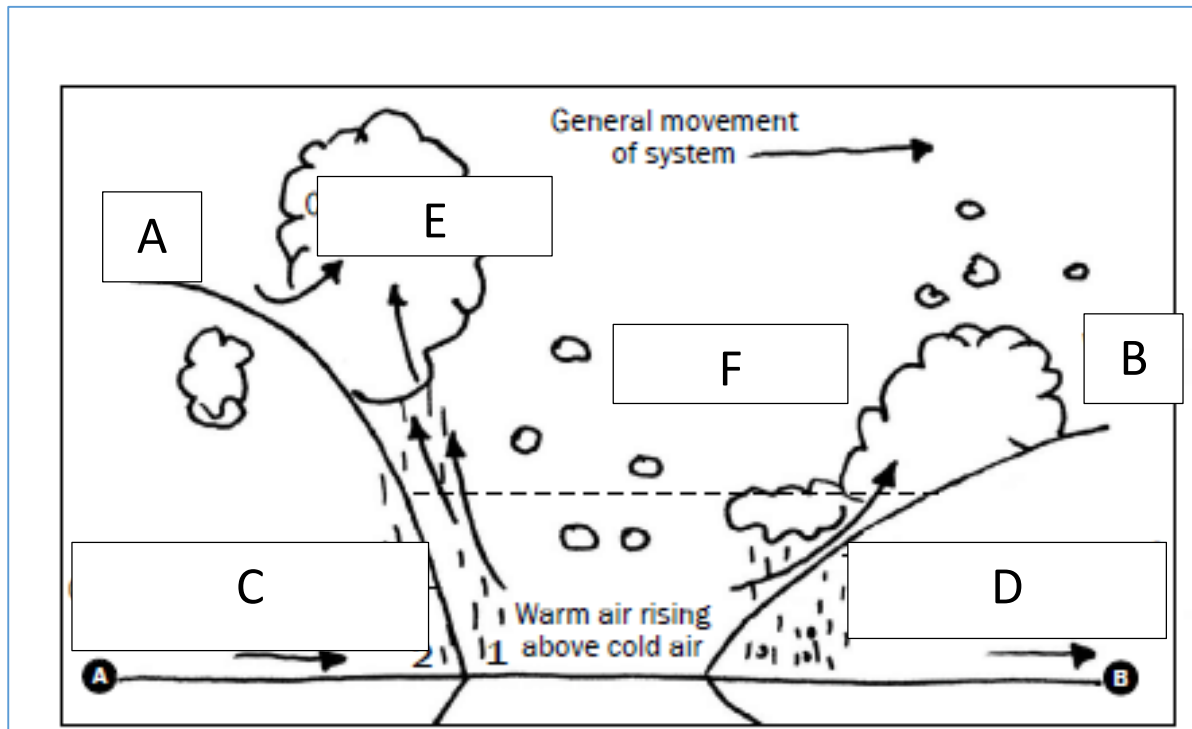
1.1. Refer to the diagram below showing a valley. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Updog



- 1.1.1. Winds that travel along the valley floors and down the slopes during the night are called (anabatic/katabatic/upslope/downslope) winds.
- 1.1.2. Temperatures at (X/Y) will be warmer during the winter months.
- 1.1.3. (R/Q) would be north-facing slopes and therefore cooler in summer.
- 1.1.4. The plantation at Q would likely be growing (delicate, exotic fruit trees/robust, hardwood timber trees).
- 1.1.5. Houses at X benefit from (the inversion layer/cultivated land) during winter.
- 1.1.6. Inversion layers are created in winter when conditions are (clear and cool/windy/cloudy/warmer than normal).
- 1.1.7. The reason the area at Y has been left barren is due to the dangers of (frost pockets/pollution domes/urban heat island effect) in winter.
- 1.1.8. If this diagram is in the Southern Hemisphere, then the river is flowing in a (westerly/easterly) direction.

(8x1) (8)

- 1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.

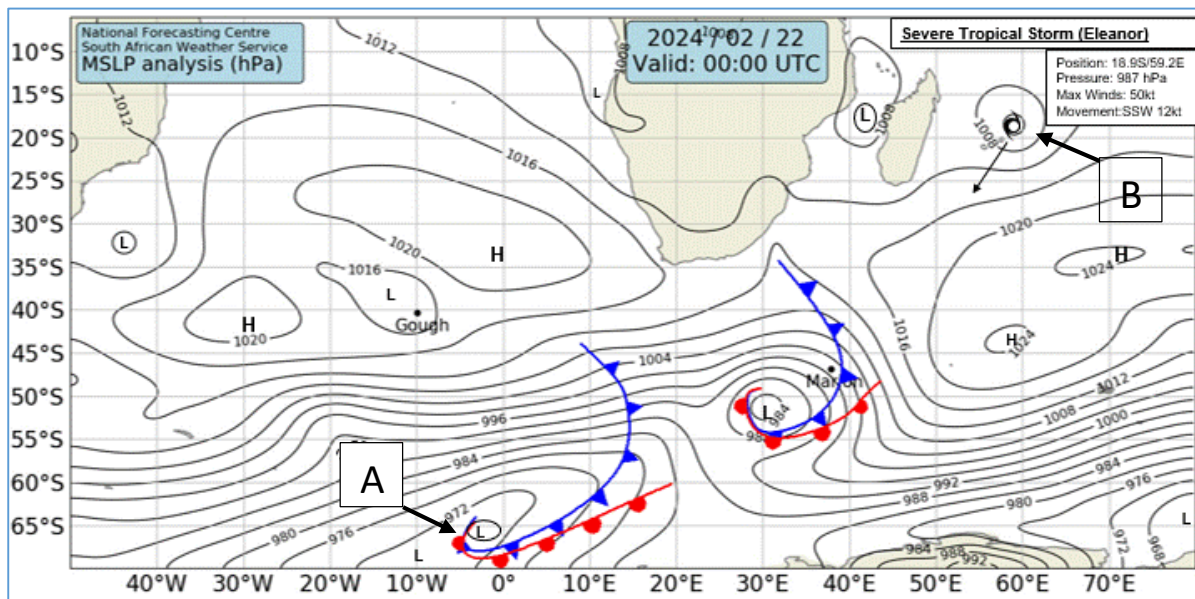


- 1.2.1. The front at A is the (warm / cold / occluded) front.
- 1.2.2. The temperatures of the air found at C, F and D are (cool, cold and warm / cold, warm and cool / warm, cold and cool) respectfully.
- 1.2.3. The symbol for the front found at B is (semi-circles / triangles / a combination of semi-circles and triangles).
- 1.2.4. The clouds that typically form at the front found at B are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. The clouds that typically form at the front found at A are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.6. The pressure found at the front labelled A is (low / high / no) pressure.
- 1.2.7. Once the front labelled A passes over a point, the pressure in that area will (increase / decrease / remain the same).

(7x1) (7)

Question 2 – Tropical Cyclones [15]

Refer to the synoptic weather map below and answer the questions that follow.



- 2.1. What season is being represented in the synoptic chart above? (1x1) (1)
- 2.2. Provide TWO pieces of evidence to justify your answer to QUESTION 2.1. (2x1) (2)
- 2.3. Name the TWO weather phenomena found at A and B respectively. (2x1) (2)
- 2.4. Tropical Storm Eleanor is not yet a mature tropical cyclone. Fully describe TWO factors necessary for the formation of mature tropical cyclones. (2x2) (4)
- 2.5. Although Tropical Storm Eleanor is not a fully-fledged cyclone, the people of Madagascar should still be on alert for the passing of this storm should its trajectory change. Describe ONE strategy that the government of Madagascar can implement to reduce the impact of this storm. (1x2) (2)
- 2.6. Tropical Storm Eleanor is predicted to miss Madagascar and head south where it will die out. Discuss why this storm will die out as it moves south. (2x2) (4)

Source(s): World Bank, the

Cities worldwide are grappling with the adverse effects of Urban Heat Island (UHI), intensified by rising global temperatures from climate change. The consequences of the UHI effect extend far beyond discomfort, with profound health and economic implications for cities and their residents.

Low-income neighbourhoods, communities lacking green spaces, and other marginalized populations bear a disproportionate burden of the economic costs associated with urban heat. Around 20% of total heat-related excess deaths globally between 2000 and 2019 occurred in East Asia, resulting in over 101,000 fatalities. Moreover, the UHI effect results in decreased productivity and higher expenses for cooling buildings, contributing to an increase in carbon emissions.

The world stands at a climate inflection point where collective action and local interventions are key. To mitigate temperature rises due to climate change, longer-term reductions in carbon emissions and international cooperation are warranted. However, UHI effects due to heat absorption by urban infrastructure and heat production by vehicles and appliances can be mitigated by local interventions. Across cities, there is an urgent need to develop sustainable solutions for mitigation and adaptation in response to UHI effects.

- 4.1. What problem is being referred to in the article? (1x1) (1)
- 4.2. Which area would likely experience the more severe conditions, the CBD or a suburban residential area? (1x1) (1)
- 4.3. List THREE factors that contribute to the problem referred to in QUESTION 4.1. (3x1) (3)
- 4.4. “Low-income neighbourhoods, communities lacking green spaces, and other marginalized populations bear a disproportionate burden of the economic costs associated with urban heat.” Explain this statement in your own words. (1x2) (2)
- 4.5. In winter urban areas typically suffer from pollution domes. In a short paragraph, briefly discuss TWO economic consequences that pollution domes have on a city and TWO viable solutions that can be implemented to mitigate these problems. (4x2) (8)

GRAND TOTAL: 60 MARKS